Electrode Shift Robust CNN for High-Density Myoelectric Pattern Recognition Control

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***Abstract*— Under the ideal condition, traditional myoelectric pattern recognition systems can achieve superior performance in recognizing motion intents. However, in practical applications, electrode shift is inevitable during the rewearing of the electrodes. The data variation caused by the shift can lead to dramatic performance degradation. In the literature, deep learning, especially convolutional neural networks (CNNs), proved effective in solving this issue. However, the aliasing effect inherent in the downsampling layer makes the common CNN shift vulnerable. The small input translations caused by the electrode shift can change the output dramatically, leading to incorrect classification. In this article, we propose a shift-robust CNN (SR-CNN) that replaces the downsampling layer with the ensemble of an anti-aliasing filter and adaptive polyphase sampling (APS) module. The anti-aliasing filter first counters the high-frequency components that can lead to unreasonable pulses during downsampling. The APS module then subsamples feature maps from several potential components adaptively, reserving the critical and stable information when the shift occurs. Experiments on two high-density surface electromyography (HD-sEMG) datasets show that the proposed SR-CNN method outperforms the state-of-the-art baselines. This study provides a promising solution to realize robust myoelectric control against the electrode shift.**

***Index Terms*— Anti-aliasing, deep learning, electrode shift, myoelectric control, pattern recognition.**

I. Introduction

ELECTROMYOGRAPHY (EMG) signal measures muscle activity in response to a nerve's stimulation of the muscle [1]. Surface EMG (sEMG) detects superficial muscles' EMG using electrodes placed over the skin, having the advantage of being noninvasive and easy to use. It is clear that people can produce repeatable sEMG signal patterns during different muscle contractions [2], [3]. These patterns can be chosen as the control source of assistive robots [4], [5], [6], [7] or rehabilitation devices [8], [9], [10], [11], and we call an sEMG-based control system a myoelectric control system (MCS). In the past few decades, beneficial to the development of the microprocessor, complex pattern recognition schemes are able to be employed in the MCS, which allows dexterous control of multiple movement patterns [7]. Specifically, the myoelectric pattern recognition (MPR) system aims to chart target movements' patterns/regularities via feature extraction and pattern recognition algorithms. With the learned regularities, the MPR system can predict a user's motion intention in real-time. A typical and effective solution was proposed by Englehart and Hudgins [12], who combined time-domain (TD) features and the linear discriminant analysis (LDA) classifier to identify hand gestures using four-channels sEMG.

Although the variety of control functions increased by MPR schemes, the robustness of the control system tends to deteriorate under nonideal situations. Some issues, for example, electrode shift [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], unknown motions [23], [24], [25], and changes of limb position and contraction strength [26], [27], [28] may significantly degrade MPR performance in real-world scenarios. Among them, electrode shift is inevitable during doffing and donning electrodes. This shift can significantly change the regularities the classifier learned before, leading to the misclassification of gestures. Ameri *et al.* [13] showed that the classification accuracies of LDA can decrease about 30% with merely a one-electrode shift. In practice, rewearing electrodes frequently occur during the user's daily life, and it is a heavy burden to retrain the classifier each time. Therefore, alleviating the interference caused by electrode shift is crucial and meaningful.

To this end, significant research efforts based on different types of electrodes have been invested. Young *et al.* [29], [30] suggested that electrodes with larger interelectrode distances can improve the performance at 1 cm shift locations, indicating that global information extracted from a larger measure area is robust to shift. Recently, high-density electrode arrays have been widely employed to collect sEMG by more and more studies. The high-density electrode array can record muscle activity in a larger area with high resolution, compared with the routine sparse electrodes. Additionally, the high-density sEMG (HD-sEMG) contains more spatial information, which has been proved to be beneficial to solve the issues of electrode shift [13], [14], [15], [16], [17], [18].

Moreover, a number of effective control schemes have been proposed. These schemes basically fall into two types—recalibrating the feature or classifier using partial data from

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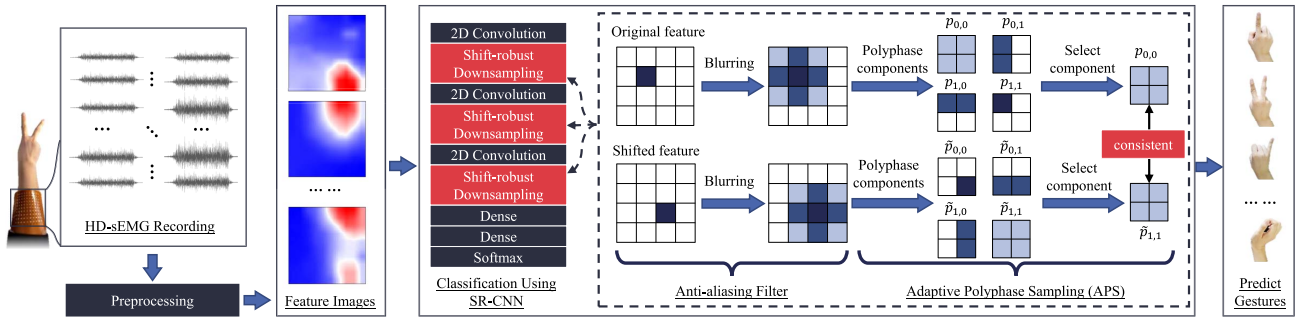


Fig. 1. Flowchart of the proposed method.

shift position [13], [15], [19], [22] and designing shift-robust control schemes [14], [17], [18], [20], [21]. With a short training session after shift, the control system can be recalibrated through transfer learning [13], increment learning [19], and shift position localization [15], [22]. These schemes can maintain a high level of classification performance after shifts, but are unavoidable to introduce extra training procedures, making the system inconvenient to use during rewearing.

A parallel line of work is designing shift-robust control schemes, which aims to build a plug-and-play MPR system in the presence of electrode shift. Wu *et al.* [14] showed that data augmentation can improve the robustness. The generated fake data simulate patterns at the shift location, hence improving the classifier's generalization. Several works [14], [17], [18], [21] also show that the spatial correlation from HD-sEMG improves the robustness. These works assume that although local features are changed during shifting, the global features, that is, spatial correlation, maintain stability.

Among the aforementioned studies, deep learning, especially convolutional neural networks (CNNs), proved effective in solving the shift problem [13], [14], [15], [16]. The capacity of CNNs to leverage spatial or temporal correlation in data is one of its most appealing features. However, all the above studies ignore the inherent aliasing effect in CNN's downsampling layer. The common CNN is shift-vulnerable, as the small input shift, or translations, can change the output dramatically, leading to incorrect classification [31]. In MPR systems, the electrode shift will cause pixel shift in the HD-sEMG feature images [14], [15], [16]. If the shifted image is input into CNNs, the unreasonable output will be produced in each downsampling layer, making the classification performance unstable.

In order to address this issue, inspired by Chaman and Dokmanić [32], we present a shift-robust CNN (SR-CNN), which is an extended version of our previous work [14]. Similarly, we first convert the HD-sEMG signals into feature images, and the electrode shift appears as the pixel shift in the image. Then, an SR-CNN is trained to extract spatial features and for classification. Compared to the routine CNN used in [14], the SR-CNN replaces the traditional downsampling layer with an ensemble of an anti-aliasing filter and an adaptive polyphase sampling (APS). The anti-aliasing filter can counter high-frequency components of the image, which can cause unreasonable pulses when downsampling the shifted image [33]. The APS subsamples a feature map from multiple sampling grids adaptively, to reserve the critical and stable information when the shift occurred. In the following sections, we will introduce the detail of the proposed method.

II. Methodology

The flowchart of the proposed method is shown in Fig. 1. The recorded HD-sEMG signals are first processed into multiple feature images. Then, the SR-CNN is trained for gesture classification. The key component in the SR-CNN is the shift-robust downsampling layer, which consists of the anti-aliasing filter and APS module. In the following sections, we will introduce each part in detail.

A. HD-sEMG Recording and Preprocessing

In this study, the high-density electrode array employed with m rows and n columns is utilized to collect HD-sEMG signals. During performing gestures, the recording device saves the raw HD-sEMG signals in the form of $m \times n \times t$, where t represents time. Given the real-time needs, that is, the system needs to recognize the gesture within 300 ms [12], we segment the HD-sEMG signals into a succession of 256-ms analysis windows with an overlap of 128 ms. The signals of relaxation are filtered by amplitude threshold manner [24]. We call each analysis window of active gestures a raw sample, with a shape of $m \times n \times 256$ ms. Then, for each channel, three features, that is, wavelength [12], f_1, and f_6 of time-dependent power spectrum descriptors (TD-PSD) [26] are extracted. Therefore, a raw sample is processed into a feature image with the shape of $m \times n \times 3$, where 3 represents three features. The previous works [14], [15] have demonstrated the effectiveness of the ensemble of these three features.

Next, as described in [14], some noises in the feature images caused by radio frequency and physiological interference are first suppressed by a median filter. Refer to the previous work [14] for details. Then, necessary interpolation and data augmentation procedures are applied to construct a training set, following the setting in [14]. Data augmentation simulates feature images at shift locations, making the classifier more generalizable. Accordingly, all these feature images constitute a training set X that includes K samples from l gesture classes: $X = \{X_i, i = 1, 2, \ldots, l\} = \{(x_k, y_k), k = 1, 2, \ldots, K\}$, where y_k labels the feature image x_k into one of the l gesture

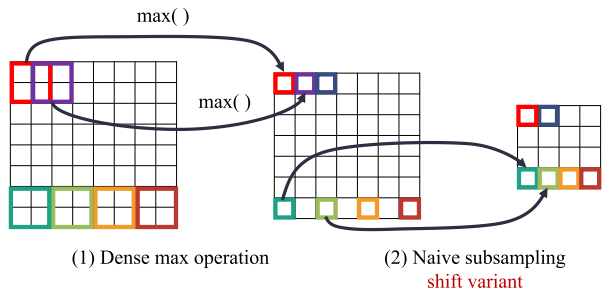


Fig. 2. Max-pooling can be divided into two parts. (1) Dense evaluation in a sliding window fashion and (2) naive subsampling.

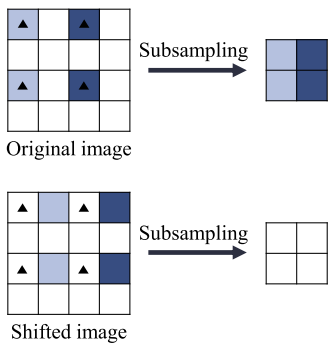


Fig. 3. Simple illustration of how subsampling impacts outputs. Subsampling saves the even component from the fixed positions (shown by black triangles) while discarding the other components. (Top) Subsampling result for an input image. (Bottom) By just shifting the input and subsequently the subsampling, an entirely different output is obtained.

classes. So, the task of gesture recognition is transformed into the image classification problem. Then, we can utilize advanced technology, such as CNNs, to dig the spatial information inherent in the HD-sEMG, which is demonstrated to be robust to electrode shift [13], [14], [15], [16].

B. Classification Using Shift-Robust CNN

1) Limitation of the Routine CNN: One of the critical reasons CNNs are sensitive to shifts in images is the downsampling or stride layers. In CNNs, downsampling expands the receptive field while reducing the parameter amount required for training. As shown in Fig. 2, the common max-pooling can be written as a composite of two parts: 1) dense evaluation in a sliding window fashion and 2) the subsequent is a naive subsampling [33]. It is also worth noting that the subsampling always chooses the even component (start at 0) while rejecting the odd one. Although ample empirical evidence suggests max-pooling provides strong task performances, the subsampling unavoidably leads to shift variation due to the ignorance of the Nyquist sampling theorem [33]. As shown in Fig. 3, the subsampling output is entirely different by just shifting the input image.

2) Anti-Aliasing Filter: Anti-aliasing or blurring the image before subsampling is the textbook answer to the aliasing problem. The image's high-frequency components are alias

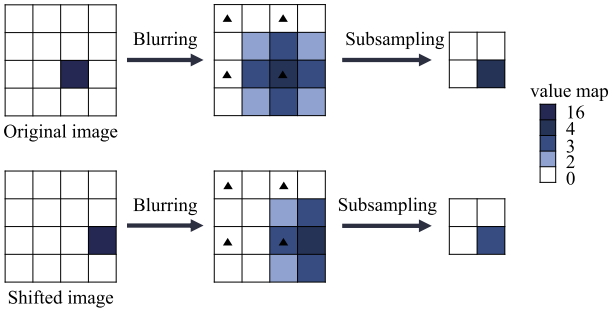


Fig. 4. Illustrative example of how anti-aliasing affects subsampling. Blurring distributes sharp elements across neighboring pixels. The resulting subsampling outputs become more similar, hence increasing the shift resilience.

into lower-frequency components if this is not done. In actuality, the blurring filter function acts as a low-pass filter, allowing the high-frequency components to be filtered out. Blurring distributes sharp elements across neighboring pixels, as demonstrated in Fig. 4. As a result, the outputs sampled throughout the grid become increasingly similar, hence increasing the shift resilience. With increasing smoothness, we test nine $m \times m$ filters ranging from 1 to 9. The filters are the outer product of 1-D Binomial filters' vectors, which give a compact approximation of the Gaussian coefficients using just integers

$$
\begin{aligned}
m = 1&: \ [1] \\
m = 2&: \ [1, 1] \\
m = 3&: \ [1, 2, 1] \\
m = 4&: \ [1, 3, 3, 1] \\
m = 5&: \ [1, 4, 6, 4, 1] \\
m = 6&: \ [1, 5, 10, 10, 5, 1] \\
m = 7&: \ [1, 6, 15, 20, 15, 6, 1] \\
m = 8&: \ [1, 7, 21, 35, 35, 21, 7, 1] \\
m = 9&: \ [1, 8, 28, 56, 70, 56, 28, 8, 1].
\end{aligned}
$$

No operation is applied when the filter size is 1, and the filter is degraded to the common average-pooling with the size of 2. Each filter is normalized before applying.

3) Adaptive Polyphase Sampling: Even though the blurring does help anti-alias, the subsampling outputs are still different due to the limitation of the nonideal low-pass filter, which cannot counter all high-frequency components completely [33]. To cope with this problem, we adopt a new downsampling method, that is, APS [32], to improve the shift robustness further. As we saw in Fig. 3, considering subsampling with stride-2, the common convolution operations only select the even components of the input images. Imagine that the subsampling could choose the odd components when the image shifted, which will lead to two same outputs. Fig. 5 illustrates the main idea of the APS. The APS first samples all alternative components and then selects one component with the highest energy.

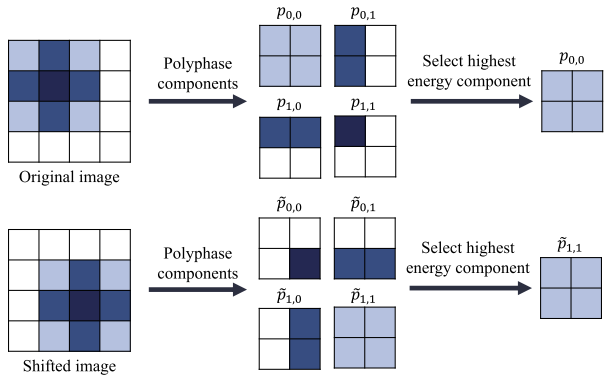


Fig. 5. Main idea of the APS. (Top) Original and (bottom) shifted images have similar polyphase components. By selecting the components with the highest energy, we can achieve the same outputs of both images.

Especially, considering the common stride-2 downsampling, the outputs can come from four possible grids. We call these potential outputs polyphase components $\{p_{ab}\}_{a,b=0}^{1}$ (see Fig. 5). The components of the 1-pixel shifted images are denoted by $\{\widetilde{p}_{ab}\}_{a,b=0}^{1}$. Fig. 5 shows that the $\{\widetilde{p}_{ab}\}_{a,b=0}^{1}$ is essentially a reordered and shifted version of the $\{p_{ab}\}_{a,b=0}^{1}$

$$\begin{aligned}\widetilde{p}_{00} &= p_{11}(n_1-1, n_2-1)\\ \widetilde{p}_{01} &= p_{10}(n_1, n_2-1)\\ \widetilde{p}_{10} &= p_{01}(n_1-1, n_2)\\ \widetilde{p}_{11} &= p_{00}(n_1, n_2).\end{aligned} \tag{1}$$

According to the above equation, the subsampling outputs of shifted images remain invariant if the identical ones are chosen from $\{p_{ab}\}_{a,b=0}^{1}$ and $\{\widetilde{p}_{ab}\}_{a,b=0}^{1}$. Following the setting in the [32], we choose the components with the highest energies, and the l_2 norm served as the metric

$$p_{\text{APS}} = p_{a_1 b_1}$$

where

$$a_1, b_1 = \arg\max_{a,b}\{\|p_{ab}\|_2\}_{a,b=0}^{1} \tag{2}$$

whereas only p_{00} is chosen as the output in the traditional max-pooling or average-pooling layers, leading to the shift variation. As shown in Fig. 1, we replace the commonly used stride-2 pooling layers with the proposed downsampling layers. In this condition, the inputted image is a feature map I with multiple channels. Accordingly, the components are defined by gathering the respective components for all channels. The overall subsampling procedure is summarized in Algorithm 1.

Algorithm 1 Shift-Robust Subsampling

1: **Input:** feature map I with multiple channels.
2: Blurring I with the anti-aliasing filter, according to Section II-B2.
3: Extracting all potential polyphase components $\{p_{ab}\}_{a,b=0}^{1}$.
4: **Output:** $p_{APS} = p_{a_1 b_1}$, according to Eq. 2.

TABLE I
STRUCTURE OF THE ADOPTED CNN

Input: HD-sEMG feature images with 3 channels
2D Convolution, 32 kernels of size (3×3), dilation is 3
Shift-robust subsampling
2D Convolution, 64 kernels of size (3×3), dilation is 3
Shift-robust subsampling
2D Convolution, 64 kernels of size (3×3), dilation is 3
Shift-robust subsampling
Fully connected layer (128 units)
Fully connected layer (6/34 units)
Softmax layer

4) Classifier Details: As shown in Table I, a CNN is trained for classification. Each convolution and fully-connected layer is followed by a ReLu activation function [34]. Multiple convolution layers are utilized to extract spatial information from HD-sEMG images $\{x_k\}_{k=1}^{K}$, and two fully-connected layers are added to predict the gesture classes $\{y_k\}_{k=1}^{K}$. The CNN architecture is selected after some pretests. This CNN is trained with the routine cross-entropy loss and optimized by the Adam algorithm [35]. The batch size is set to 32. We first train the model for 20 epochs with a learning rate of 0.001 and then fine-tune it for another 20 epochs with a small learning rate of 0.0001. Besides, to prevent overfitting, an $L2$ regularization with the weight of 0.0001 is added to all parameters. We implement all algorithms using Python language and Pytorch architecture and run them on a PC with an Intel Xeon Silver CPU and an NVIDIA GeForce RTX 3080 GPU.

C. Comparison Methods

In this study, classification accuracy is used as the performance metric. Eight comparison methods are implemented to evaluate the effectiveness of the adopted method.

LDA [12] method is popular due to its superior performance in myoelectric control and ease of implementation. For implementing this method, we first flatten the feature image x_k as a vector. Then, the LDA classifier is trained and combined with the corresponding label y_k.

SVM [36] methods use different support vector machines (SVMs) to classify the gestures. The SVM classifier is widely used in various fields. As in the LDA method, we first flatten the feature images and subsequently implement three SVM methods with different kernel types to classify the feature vectors. The three kernel types are linear, polynomial, and radial basis function (RBF). Accordingly, the three SVM classifiers are denoted as **SVM-linear**, **SVM-polynomial**, **and SVM-RBF**.

iGLCM [18] method combines improved gray-level co-occurrence matrix (iGLCM) and the LDA classifier to recognize gestures. The iGLCM can extract spatial information among the HD-sEMG signals. It has been proved to improve the robustness of the myoelectric system against electrode shift. Following the parameter setting in [18], the quantization level is 16. The input data are feature images extracted in Section II-A.

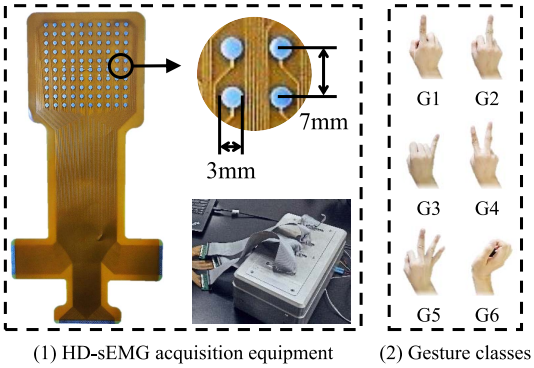


Fig. 6. (1) Experimental device and (2) gestures adopted in the Custom Dataset.

TSD-LDA [37] method combines the temporal–spatial descriptors (TSDs) and the LDA classifier for identifying gestures. The TSD features extract spatial information from combined sEMG channels. As mentioned above, the spatial information is propitious to alleviate the effect caused by the electrode shift.

The **CNN** [14], [38] method is a simple version of the proposed method. This method only consists of typical neural network layers, including convolutional layers, average-pooling layers, fully connected layers, and so on.

The **AUG-DCNN** [14] method adopts a dilated CNN (DCNN) for better extracting spatial information from HD-sEMG. Besides, a data augmentation procedure is applied to improve the generalization of the DCNN model.

D. Statistics Analysis

One-way repeated-measures ANOVAs are conducted to compare all methods (nine methods: LDA, SVM-linear, SVM-polynomial, SVM-RBF, iGLCM, TSD-LDA, CNN, AUG-DCNN, and the proposed SR-CNN method). Post hoc multiple comparisons are performed with LSD corrections. The significant level is set to 0.05. All analyses are performed using SPSS software (ver. 26, SPSS Inc., Chicago, IL, USA).

III. Experiments and Results

A. Dataset

Two datasets were used to evaluate the proposed method. The first dataset was collected in our previous works, and we call it the Custom Dataset [14], [15]. Another dataset was the PR Dataset recently published in [38]. They both collected HD-sEMG signals from different sessions, which will introduce the electrode shift issue during myoelectric control. Therefore, they are well suited to verify the effectiveness of the proposed method. We will describe these two datasets below.

1) Custom Dataset: This dataset collected HD-sEMG signals of six gestures from ten intact subjects (six males and four females, aged 24–35 years). The acquisition system and the adopted gestures are shown in Fig. 6. The electrode array contains 100 monopolar electrodes arranged in the form of a 10×10 matrix. The sample rate is 1000 Hz.

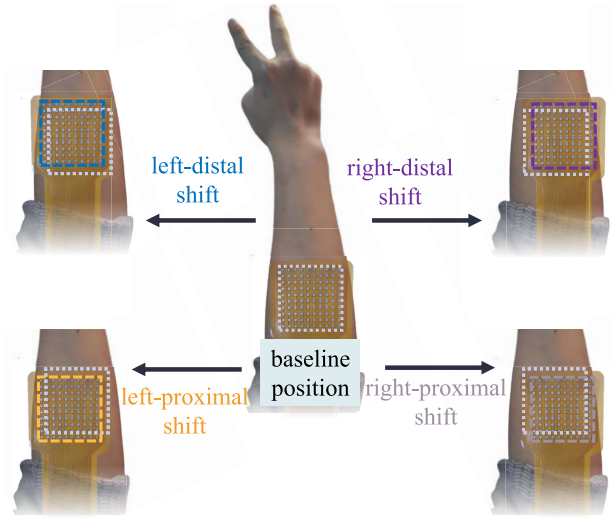


Fig. 7. Examples of electrode shifts adopted in the Custom Dataset, where the gray square represents the baseline position. The electrode array is shifted by approximately 1 cm in four directions, respectively.

TABLE II

Classification Accuracies' Average Across All Subjects in Both Datasets When Electrode Arrays Are **Not Shifted**. The Bold Values Represent the Best Results

Method	Custom Dataset	PR Dataset
LDA [12]	99.78%	82.35%
SVM-linear [36]	99.70%	85.41%
SVM-polynomial [36]	99.53%	85.14%
SVM-RBF [36]	99.64%	79.98%
iGLCM [18]	99.57%	68.02%
TSD-LDA [37]	**99.91**%	-
CNN [14], [38]	99.48%	89.84%
AUG-DCNN [14]	99.54%	89.65%
SR-CNN	99.57%	**90.37**%

For each subject, the electrode array was put at different locations to investigate the effect of electrode shift. The array was first placed on the skin's surface over the extensor digitorum and extensor carpi ulnaris, and this position was denoted as the baseline position. Then, the electrode array was shifted in four diagonal directions, that is, left-distal, right-distal, left-proximal, and right-proximal. The shift distance was approximately 1 cm. Fig. 7 shows the baseline position and four shift positions. In addition, a random shift was implemented to simulate the practical usage, where the range of this shift did not exceed those above four designated shifts. In summary, for each subject, the dataset recorded HD-sEMG signals from six electrode positions.

For each position, the subjects were asked to perform five repetitions of each of six gestures [see Fig. 6(2)] with isometric muscle contractions at a consistently moderate force. Each repeated gesture was maintained for about 5 s. The six gestures were index finger extension (G1), middle finger extension (G2), little finger extension (G3), the extension of both index and middle fingers (G4), the extension of the last three fingers (G5), and wrist extension (G6). Refer to [14] for more details regarding this dataset.

TABLE III

CLASSIFICATION ACCURACIES' AVERAGE ACROSS FIVE SHIFT POSITIONS ON THE CUSTOM DATASET. THE BOLD VALUES REPRESENT THE BEST RESULTS

Method \ Subject	1	2	3	4	5	6	7	8	9	10	**Average**
LDA [12]	79.56%	81.40%	88.55%	80.68%	80.94%	79.40%	69.55%	65.00%	82.23%	77.29%	78.46%
SVM-linear [36]	90.59%	91.60%	93.92%	83.31%	86.44%	90.33%	84.24%	87.69%	78.65%	85.17%	87.19%
SVM-polynomial [36]	91.97%	88.40%	94.55%	82.82%	85.11%	90.18%	85.04%	84.49%	81.24%	85.34%	86.91%
SVM-RBF [36]	92.92%	94.72%	92.59%	86.52%	86.67%	92.46%	86.12%	88.20%	87.57%	86.43%	89.42%
iGLCM [18]	84.40%	88.82%	95.63%	86.51%	84.74%	81.76%	75.72%	87.32%	83.83%	83.95%	85.27%
TSD-LDA [37]	97.08%	96.98%	86.33%	90.87%	86.34%	87.12%	90.13%	81.73%	86.29%	84.60%	88.75%
CNN [14]	89.63%	94.31%	94.88%	81.97%	80.97%	88.04%	83.00%	85.82%	78.66%	83.58%	86.09%
AUG-DCNN [14]	99.05%	95.58%	98.04%	92.65%	**94.72%**	95.89%	90.68%	96.06%	**93.92%**	96.81%	95.34%
SR-CNN	**99.26%**	**99.12%**	**98.99%**	**97.21%**	94.60%	**96.98%**	**92.76%**	**96.84%**	92.23%	**97.60%**	**96.56%**

2) PR Dataset: PR Dataset is the subdataset of the Hyser Dataset [38]. This dataset adopted four electrode arrays to collect HD-sEMG signals from the forearm extensor and flexor muscles. Each electrode array contains 64 gelled elliptical electrodes, arranged in an 8×8 electrode layout. The sample rate is 2048 Hz.

Twenty intact subjects (12 males, eight females, aged 21-34 years) were asked to perform two repetitions for each of the 34 hand gestures. Each repeated gesture was maintained for a 4-s duration. In addition, each subject's sEMG data were obtained twice on different days using the same experimental procedure. Therefore, the electrode position will be shifted in two acquisitions. Refer to [38] for more details regarding this dataset.

B. Classification Performance Under Different Conditions

1) Electrode Nonshift: In this case, training and testing data are collected while the electrode arrays are placed at the same location. For the Custom Dataset, only the sEMG data from the baseline position are used for model evaluation. Specifically, we utilize a fivefold cross-validation strategy where four repetitions of the raw sEMG data for each gesture are used for training, and the remaining one repetition is used for testing. This process will be iterated five times until all data are used for training and testing. The subject-dependent strategy was adopted for cross-validation. The final classification accuracy is the average of the accuracies from the five testing folds.

For the PR Dataset, following the setting in [38], we pooled data of 20 subjects from one day in one session to expand the amount of the data. Then, for sEMG data from each session, a twofold cross-validation strategy is used for model evaluation. Each fold contains one repetition of the sEMG data of all gestures. Therefore, the electrode position is the same for the two folds. Each time we used data from onefold as a testing set and data from another fold as a training set. Finally, the classification accuracy is the average over cross-validation folds.

Table II shows classification accuracies when the electrode arrays are not shifted across training and testing phases. For the Custom Dataset and PR Dataset, the filter sizes of the SR-CNN are set to 7 and 1, respectively. For the Custom Dataset, all methods achieved nearly 100% accuracies, and no statistically significant difference was observed in these results ($p > 0.05$). For the PR Dataset, deep-learning-based methods, namely CNN, AUG-DCNN, and SR-CNN, achieved comparable performance, and the accuracies are significantly higher than other traditional methods. Note that the TSD-LDA result was not given because the dimensionality of the TSD feature extracted from the PR Dataset was too high (dimensionality reached 230k), and our server cannot load and process all subjects' feature data at once.

TABLE IV

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS UNDER THE ELECTRODE SHIFT CONDITION ON THE PR DATASET. THE BOLD VALUES REPRESENT THE BEST RESULTS

Method	Training session: 2 Testing session: 1	Training session: 1 Testing session: 2	**Average**
LDA [12]	44.47%	44.05%	44.26%
SVM-linear [36]	58.07%	57.39%	57.73%
SVM-polynomial [36]	60.68%	61.50%	61.09%
SVM-RBF [36]	58.50%	60.08%	59.29%
iGLCM [18]	50.11%	50.54%	50.33%
CNN [38]	61.20%	61.30%	61.25%
AUG-DCNN [14]	69.20%	69.91%	69.55%
SR-CNN	**70.42%**	**71.51%**	**70.97%**

2) Electrode Shift: In this case, training and testing data are collected while the electrode arrays are placed at different locations. For the Custom Dataset, all five repetitions' sEMG data from the baseline position were used for training, while all data from five shift positions were used for testing. For this dataset, the filter size of the SR-CNN is set to 7. Table III summarizes the classification performance obtained by the proposed method and eight comparison methods. It is worth noting that the proposed SR-CNN method achieves the highest average accuracy of 96.56%. The repeated-measures ANOVA indicates that the proposed method is significantly better than other methods ($p < 0.05$ for the proposed method vs. any other method).

For the PR Dataset, we used data collected on different days for model training and testing, respectively. Specifically, following the setting in [38], we first pooled data of all subjects from one day into one session. Therefore, two sessions' sEMG data were obtained. Then, the data from the two sessions were alternately used as the training set, while the data from the other session were used as the testing set accordingly. For this dataset, the filter size of the SR-CNN is set to 1. Table IV presents the classification accuracies of different methods on the PR Dataset. Note that the TSD-LDA result was not given because the dimensionality of the TSD feature

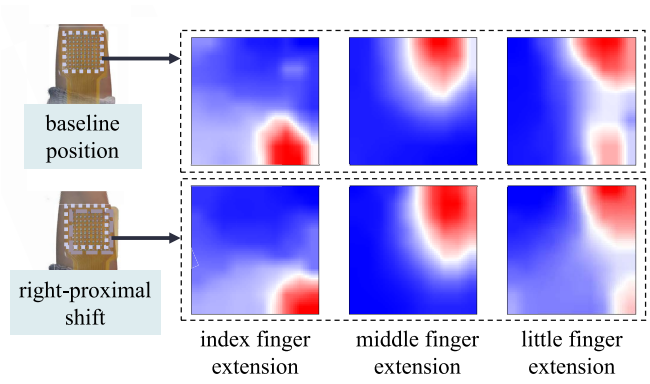


Fig. 8. Representative examples of three gestures' feature images from one subject in the Custom Dataset when the electrode array is placed at the baseline position (top) and the right-proximal shift (bottom). Only the wavelength feature is plotted for simplicity.

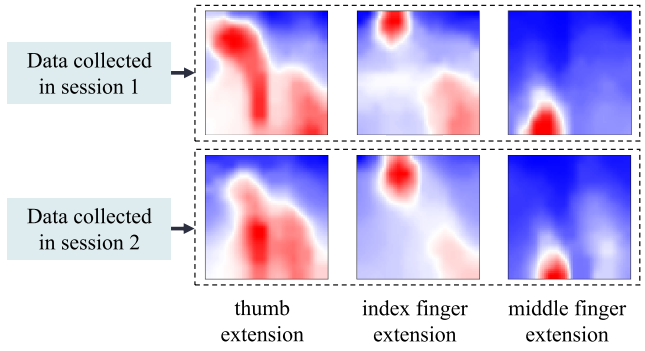


Fig. 9. Representative examples of three gestures' feature images from one subject in the PR Dataset when the data are collected in session 1 (top) and session 2 (bottom). Only the wavelength feature is plotted for simplicity.

from the PR Dataset was too high (about 230k) to process. The results indicate that the proposed method outperforms other comparison methods under different testing conditions.

C. Examples of Feature Images of HD-sEMG

Figs. 8 and 9 show the representative examples of feature images with the array placed at different positions. Note that different gestures have distinct patterns, while one gesture has similar patterns across different electrode positions. Accordingly, we transform the problem of gesture recognition into an image classification problem, where the electrode shift causes the pixel shift in the input image. Then, this study aims to improve the generalization ability of the classifier so that it can still recognize gestures well when the pixels are shifted.

D. Experiments With Blurring Filters of Different Sizes

As described in Section II-B2, the anti-aliasing filter plays an important role in resisting the influence of shifts. Here, we investigate the effect of different sizes of blurring filters on the classification results. The SR-CNN models with different filter sizes are trained and evaluated on the Custom Dataset. Fig. 10 presents the classification accuracies obtained under the electrode shift conditions. Compared with no filter involved

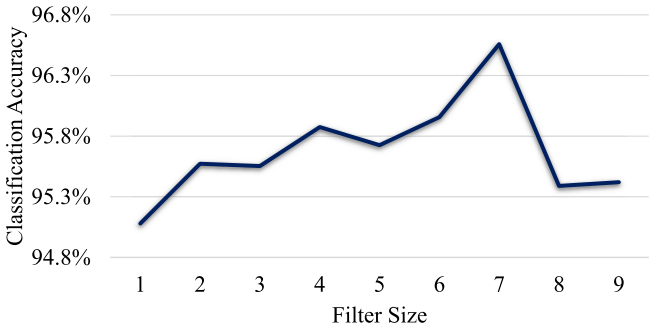


Fig. 10. Classification accuracies were achieved by the **SR-CNN** method using different blurring filters. Each point is average from all shift conditions.

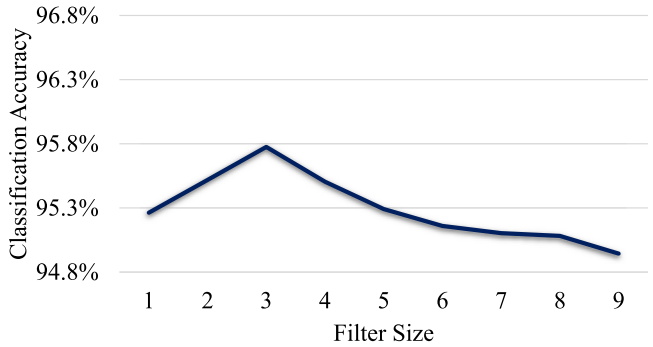


Fig. 11. Classification accuracies were achieved by the **baseline + blur** method using different sizes of blurring filters. Each point is average from all shift positions.

TABLE V

RESULTS OF THE ABLATION STUDY. THE CLASSIFICATION ACCURACIES ARE AVERAGED FROM THE RESULTS OF ALL SUBJECTS IN THE FIVE SHIFT POSITIONS. THE BOLD VALUE REPRESENTS THE BEST RESULT

Method	Accuracy
baseline	95.26%
baseline+blur	95.78%
baseline+APS	95.08%
baseline+blur+APS (**SR-CNN**)	**96.56%**

(filter size is 1), an extensive range of filter sizes can significantly improve the classification performance. Especially, the accuracy reaches 96.55% when the size is 7.

E. Ablation Study

In this study, the procedure of shift-robust downsampling consists of two parts: one is the anti-aliasing filter, and the other is the APS. To evaluate the effect of each part, we perform ablation experiments on the Custom Dataset. We first construct a **baseline** method, which replaces all shift-robust downsampling layers in Table I with the common average-pooling layers. Then, we add an anti-aliasing filter before each average-pooling layer. We denote this method as **baseline + blur**. Fig. 11 shows the classification performances averaged across all shift positions, with the increase in filter size. We can notice that the best accuracy is 95.78% obtained when the filter size is 3. Next, we replace all average-pooling layers in the baseline method with the APS module, and we denote this method as **baseline + APS**. The comparison results of the

aforementioned methods are summarized in Table V. It should be noted that only the proposed SR-CNN method, that is, the combination of the anti-aliasing filter and the APS module, can achieve the best accuracy.

IV. Discussion

In the application of myoelectric control, it is crucial to counter the issue caused by electrode shift. In this article, we propose a shift robust CNN, which replaces the common downsampling layer with the combination of an anti-aliasing filter and APS, making the classifier robust to the electrode shift.

The classification accuracies on both datasets reveal that although most methods can achieve high accuracies under the nonshift condition, the classification performance cannot be generalized to the shift positions well. As an example, the LDA method is able to achieve a 99.78% classification accuracy on the Custom Dataset in the nonshift case, while in the electrode shift case, only a 78.46% accuracy is achieved. This can be attributed to the sensitivity of the LDA classifier, which identifies each image according to the pattern represented through all pixels. However, the electrode shift changes all pixel values within the images, and these cumulative changes will cause the current image pattern to conflict with the regularity of the LDA learned before, leading to the misclassification of the LDA.

To alleviate the changes brought by the shift, many studies dedicated to extracting spatial correlations have become popular in the field of myoelectric control [14], [17], [18], [21], [37]. They suppose that although pixels in feature images are changed during shifting the electrode, the correlation between channels is stable. That is to say, even though images are from different positions, they are quite similar with almost the same highlighted region, implying the same muscle components under the skin surface. In this study, we select the iGLCM [18] and TSD [37] as representative methods for extracting spatial correlations inherent in the HD-sEMG signals. The classification results of the shift positions on the two datasets indicate that both methods outperform the routine LDA methods, but the performances are comparable to the SVM methods. The reason may be that they are usually sensitive to only certain spatial attributes, such as texture information [18], resulting in incomplete extraction of spatial information and thus unsatisfactory classification performances.

Given that CNN has revolutionized the field of computer versions, it is natural to adopt CNN to extract spatial information from HD-sEMG images. The AUG-DCNN method [14] adopts the CNN to extract spatial information and, at the same time, applies data augmentation to improve the robustness against the electrode shift. Even though this method has achieved high accuracies on the two datasets, there is still room for improvement. A key point often overlooked in previous studies is that the CNN itself is vulnerable to pixel shift of the input image.

As stated in Section II-B1, the common max-pooling layer in CNNs can be divided into two parts: the first part is dense evaluation in a sliding window fashion, and the subsequent part is a simple subsampling, where the subsampling only retains the first component and discards the others. It should be noted that the outputs from simple subsampling change dramatically when the input image shifts. This phenomenon can be explained by the Nyquist sampling theorem, where an inappropriate sampling (subsampling) will alias the output, making the results shift vulnerable. Therefore, one straightforward solution is applying a blurring filter before each subsampling. The blurring filter can suppress the high-frequency components without the loss of some important details, such as muscle locations. The ablation experiments design a baseline + blur method, which adds a blurring filter before each average-pooling layer. The classification result of the baseline + blur method shows that, compared with the traditional baseline method, the baseline + blur method produces a better result when the size is set to 3, indicating the effectiveness of the anti-aliasing filter. However, the classification accuracy of the baseline + blur method decreases with the further increase of the filter size. This can be attributed to the fact that although the blurring filter reduces the adverse effects of the shift, it also dilutes the key information. If the two images are severely blurred, that is, blurred with a large filter, it will not be easy to extract distinct information and classify them. Specifically, if the pooling layer selects the key information, higher recognition performance can be obtained, and if the important information falls in a position other than the first component, the information will be discarded, which may affect the final classification result.

To further improve the robustness against electrode shifts, we combine the APS module with the blurring filter. The APS can adaptively select the components with the highest energy. Given that the feature image is blurred by the filter, its global spatial information becomes stable. That is to say, the component of the highest energy may be relatively stable across shifts [32]. The classification results indicate that the proposed method can achieve state-of-the-art performances on the two datasets.

In addition, the average testing time for the models used in the two datasets is 2 ms (Custom Dataset) and 1 ms (PR Dataset) per sample, respectively. The testing time is short enough to meet the real-time control requirements. Therefore, it is promising to transition the proposed method to real applications, such as prosthetics and exoskeletons.

There are limitations to this study. The current study adopts the fixed size of blurring filters across all downsampling layers. In the network, different layers can represent different scales of spatial information. Therefore, different sizes of the filter could be applied to different layers for better anti-aliasing. Ongoing work is looking at automatically determining the suitable filter size for each layer. Additionally, the investigation into using multiscale technology to counter the shift issue is ongoing. It is hoped that the low-scale images are stable during shifting, while the images with high resolution can maintain rich details for classification. The fusion of the multiscale information may produce a robust classifier.

V. Conclusion

This study proposed an SR-CNN for alleviating the interference caused by electrode shift. The proposed method replaced

the common downsampling layer with an ensemble of an anti-aliasing filter and an APS. The proposed method obtained state-of-the-art performances on two datasets, indicating that the SR-CNN method was a promising solution to realize robust myoelectric control against the electrode shift.

REFERENCES

[1] Z. Wang *et al.*, "Motion analysis of deadlift for trainers with different levels based on body sensor network," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–12, 2021.

[2] H. Xu and A. Xiong, "Advances and disturbances in sEMG-based intentions and movements recognition: A review," *IEEE Sensors J.*, vol. 21, no. 12, pp. 13019–13028, Jun. 2021.

[3] T. Bao, S. A. R. Zaidi, S. Xie, P. Yang, and Z.-Q. Zhang, "A CNN-LSTM hybrid model for wrist kinematics estimation using surface electromyography," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

[4] J. M. Hahne, M. A. Schweisfurth, M. Koppe, and D. Farina, "Simultaneous control of multiple functions of bionic hand prostheses: Performance and robustness in end users," *Sci. Robot.*, vol. 3, no. 19, Jun. 2018, Art. no. eaat3630.

[5] R.-E. Precup, T.-A. Teban, A. Albu, A.-B. Borlea, I. A. Zamfirache, and E. M. Petriu, "Evolving fuzzy models for prosthetic hand myoelectric-based control," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 7, pp. 4625–4636, Jul. 2020.

[6] A. Moin *et al.*, "A wearable biosensing system with in-sensor adaptive machine learning for hand gesture recognition," *Nature Electron.*, vol. 4, pp. 54–63, Jan. 2021.

[7] A. Furui *et al.*, "A myoelectric prosthetic hand with muscle synergy–based motion determination and impedance model–based biomimetic control," *Sci. Robot.*, vol. 4, no. 31, Jun. 2019, Art. no. eaaw6339.

[8] P. Qin and X. Shi, "A novel method for lower limb joint angle estimation based on sEMG signal," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021.

[9] S. Pasinetti, M. Lancini, I. Bodini, and F. A. Docchio, "A novel algorithm for EMG signal processing and muscle timing measurement," *IEEE Trans. Instrum. Meas.*, vol. 64, no. 11, pp. 2995–3004, Nov. 2015.

[10] D. Leonardis *et al.*, "An EMG-controlled robotic hand exoskeleton for bilateral rehabilitation," *IEEE Trans. Haptics*, vol. 8, no. 2, pp. 140–151, Jun. 2015.

[11] F. B. Wagner *et al.*, "Targeted neurotechnology restores walking in humans with spinal cord injury," *Nature*, vol. 563, no. 7729, pp. 65–71, 2018.

[12] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, Jun. 2003.

[13] A. Ameri, M. A. Akhaee, E. Scheme, and K. Englehart, "A deep transfer learning approach to reducing the effect of electrode shift in EMG pattern recognition-based control," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 2, pp. 370–379, Feb. 2020.

[14] L. Wu, X. Zhang, K. Wang, X. Chen, and X. Chen, "Improved high-density myoelectric pattern recognition control against electrode shift using data augmentation and dilated convolutional neural network," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 12, pp. 2637–2646, Dec. 2020.

[15] X. Zhang, L. Wu, B. Yu, X. Chen, and X. Chen, "Adaptive calibration of electrode array shifts enables robust myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 67, no. 7, pp. 1947–1957, Jul. 2020.

[16] R. Hu, X. Chen, X. Zhang, and X. Chen, "Adaptive electrode calibration method based on muscle core activation regions and its application in myoelectric pattern recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 11–20, 2021.

[17] L. Pan, D. Zhang, N. Jiang, X. Sheng, and X. Zhu, "Improving robustness against electrode shift of high density EMG for myoelectric control through common spatial patterns," *J. NeuroEng. Rehabil.*, vol. 12, no. 1, pp. 1–16, 2015.

[18] J. He and X. Zhu, "Combining improved gray-level co-occurrence matrix with high density grid for myoelectric control robustness to electrode shift," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 9, pp. 1539–1548, Sep. 2016.

[19] Q. Ding, X. Zhao, J. Han, C. Bu, and C. Wu, "Adaptive hybrid classifier for myoelectric pattern recognition against the interferences of outlier motion, muscle fatigue, and electrode doffing," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 27, no. 5, pp. 1071–1080, May 2019.

[20] L. Hargrove, K. Englehart, and B. Hudgins, "A training strategy to reduce classification degradation due to electrode displacements in pattern recognition based myoelectric control," *Biomed. Signal Process. control*, vol. 3, pp. 175–180, Apr. 2008.

[21] A. Stango, F. Negro, and D. Farina, "Spatial correlation of high density EMG signals provides features robust to electrode number and shift in pattern recognition for myocontrol," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 23, no. 2, pp. 189–198, Mar. 2015.

[22] J. He, X. Sheng, X. Zhu, and N. Jiang, "A novel framework based on position verification for robust myoelectric control against sensor shift," *IEEE Sensors J.*, vol. 19, no. 21, pp. 9859–9868, Nov. 2019.

[23] Y. Lin, R. Palaniappan, P. D. Wilde, and L. Li, "Reliability analysis for finger movement recognition with raw electromyographic signal by evidential convolutional networks," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 96–107, 2022.

[24] E. J. Scheme, B. S. Hudgins, and K. B. Englehart, "Confidence-based rejection for improved pattern recognition myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 60, no. 6, pp. 1563–1570, Jun. 2013.

[25] J. W. Robertson, E. J. Scheme, and K. B. Englehart, "Effects of confidence-based rejection on usability and error in pattern recognition-based myoelectric control," *IEEE J. Biomed. Health Informat.*, vol. 23, no. 5, pp. 2002–2008, Sep. 2019.

[26] A. Al-Timemy, R. Khushaba, G. Bugmann, and J. Escudero, "Improving the performance against force variation of EMG controlled multifunctional upper-limb prostheses for transradial amputees," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 24, no. 6, pp. 650–661, Jun. 2016.

[27] E. Campbell, A. Phinyomark, and E. Scheme, "Current trends and confounding factors in myoelectric control: Limb position and contraction intensity," *Sensors*, vol. 20, no. 6, p. 1613, Mar. 2020.

[28] Y. Teh and L. J. Hargrove, "Understanding limb position and external load effects on real-time pattern recognition control in amputees," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 28, no. 7, pp. 1605–1613, Jul. 2020.

[29] A. J. Young, L. J. Hargrove, and T. A. Kuiken, "The effects of electrode size and orientation on the sensitivity of myoelectric pattern recognition systems to electrode shift," *IEEE Trans. Biomed. Eng.*, vol. 58, no. 9, pp. 2537–2544, Sep. 2011.

[30] A. J. Young, L. J. Hargrove, and T. A. Kuiken, "Improving myoelectric pattern recognition robustness to electrode shift by changing interelectrode distance and electrode configuration," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 3, pp. 645–652, Mar. 2012.

[31] A. Azulay and Y. Weiss, "Why do deep convolutional networks generalize so poorly to small image transformations?" *J. Mach. Learn. Res.*, vol. 20, no. 184, pp. 1–25, 2019.

[32] A. Chaman and I. Dokmanic, "Truly shift-invariant convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 3773–3783.

[33] R. Zhang, "Making convolutional networks shift-invariant again," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7324–7334.

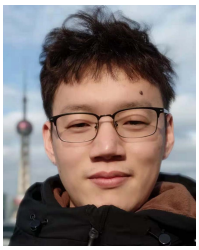
[34] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines," in *Proc. 27th Int. Conf. Int. Conf. Mach. Learn. (ICML)*. Madison, WI, USA: Omnipress, 2010, pp. 807–814.

[35] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent., (ICLR)*, Y. Bengio and Y. LeCun, Eds. San Diego, CA, USA, May 2015.

[36] C. C. Chang and C. J. Lin, "LIBSVM: A library for support vector machines," *ACM Trans. Intell. Syst. Technol.*, vol. 2, no. 3, pp. 1–27, 2011.

[37] R. N. Khushaba, A. H. Al-Timemy, A. Al-Ani, and A. Al-Jumaily, "A framework of temporal-spatial descriptors-based feature extraction for improved myoelectric pattern recognition," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 25, no. 10, pp. 1821–1831, Oct. 2017.

[38] X. Jiang *et al.*, "Open access dataset, toolbox and benchmark processing results of high-density surface electromyogram recordings," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 1035–1046, 2021.



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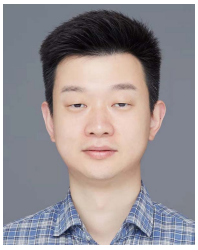

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